



भारतीय प्रौद्योगिकी संस्थान कानपुर
Indian Institute of Technology Kanpur



**Department of Chemical
Engineering**

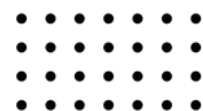
Class of 2026-27



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Message from HOD



**Dr. Nitin Kaistha,
Professor and Head**

Ranked among the nation's top schools in Chemical Engineering, Chemical Engineering at IITK offers arguably the most rigorous undergraduate and graduate programs to prepare competent and confident graduates, ready to take up challenging professional roles in the industry. Our core strengths are the nation's best students enrolling in our programs, distinguished faculty of international standing and state-of-the-art research and pedagogical infrastructure. Their juxtaposition fosters an academic environment that nurtures higher order thinking skills, creativity and innovation beyond traditional boundaries. We strongly believe that the rigorous academic and vibrant cultural ambiance of IITK shapes our graduates to be amongst the best in the world, both in terms of competence and technical skills as well as a well-rounded personality. Naturally, they are highly sought after by a very diverse portfolio of recruiters.

As the next academic year begins, we invite you to actively participate in the recruitment season to pick-up some of the best talents in the country, who will be graduating in summer 2027, for your organization.

Vision & Mission



Vision :

We strive to become the leading contributor to India's technological ambitions and societal needs by rigorously preparing our students to be leaders in these efforts, and by the creation and dissemination of knowledge at the highest levels.

Mission:

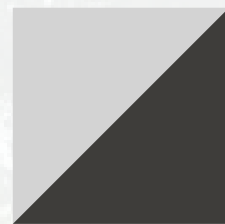
To produce leaders and engineers who can solve problems with disruptive ideas and create a new path for future generation of chemical engineers.

Provide vibrant research programs to attract top talents and undertake grand challenge problems.

Strengthen industry-academia relationships, solve industry and society's needs through disruptive technologies, and shape engineering science and technology through an interdisciplinary approach.



Courses



Elective Courses :

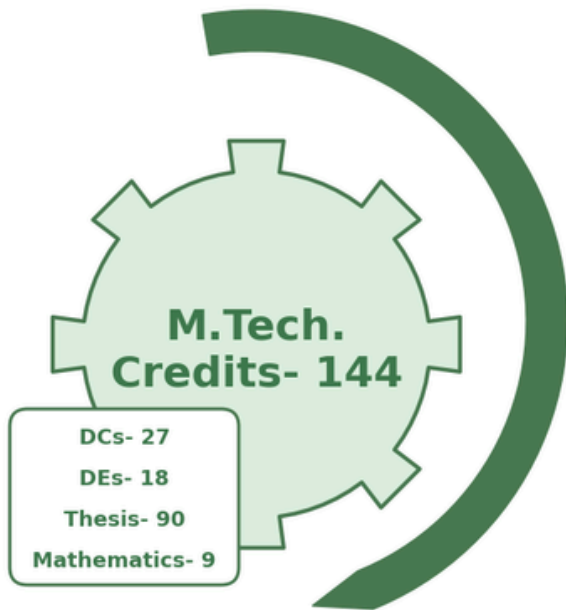
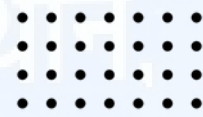
- Petroleum Refinery Engineering
- Computer Aided Process Control
- Molecular Modelling & Simulation
- Machine Learning in Chemical Engineering
- The Structure and Rheology of Complex Fluids Process Engineering & Optimization
- Chemical Plant Safety & Hazard
- Assessment Reaction Engineering of Polymers
- Advanced Fluid Mechanics
- Mesoscale & Continuum Simulations
- Mechanics of Soft Matter
- Statistical Thermodynamics Nano-sciences & Micro-fluids
- Hydrodynamic Stability
- Process Engineering Principles in Microfabrication
- Introduction to Electrochemical Engineering
- Data Science For Chemical Engineers
- Artificial Intelligence In System Biology

Core Courses :

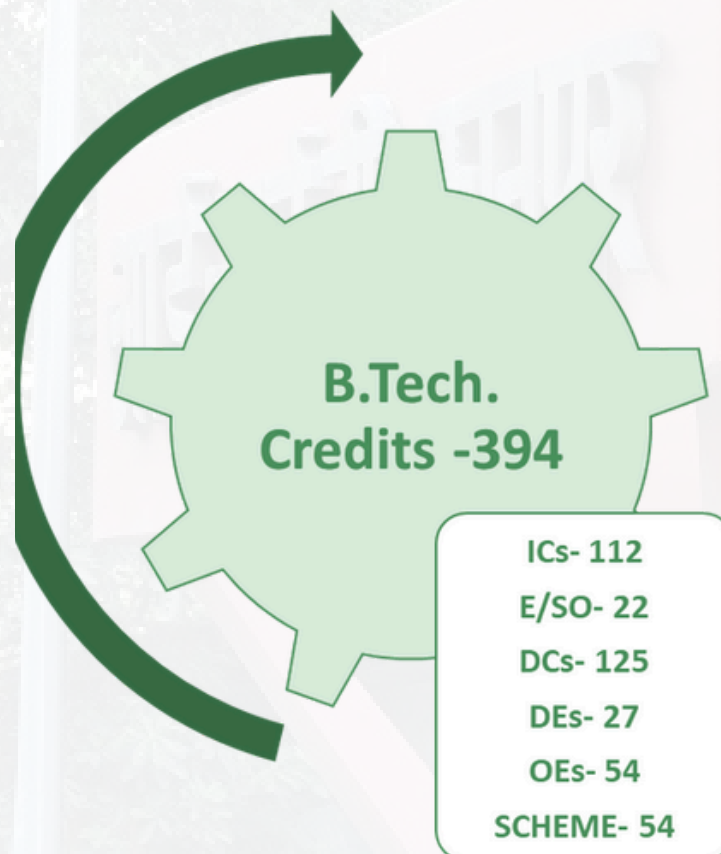
- Thermodynamics
- Fluid Mechanics & its Application
- Heat Transfer & its Applications
- Mass Transfer & its Application
- Chemical Process Industries
- Process Dynamics and Control
- Chemical Reaction Engineering
- Biochemical Engineering
- Transport Phenomena
- Applied Numerical Methods in Engineering
- Mathematical Methods In Chemical Engineering
- Chemical Process Simulation
- Chemical Engineering Design



Credits Required



DCs- Department Compulsory
DEs- Department Electives
OEs- Open Electives
IC- Institute Core
E/SO- Engineering Science Courses
SCHEME- Humanities and Social Sciences
(HSS) Courses



Sectors We Add Value



- Refining and Petrochemicals
- Process Design and Engineering Consultancy
- Fertilizers and Agrochemicals
- Oil & Gas Exploration and Production
- Coal and Mineral Processing
- Pharmaceuticals and Biotech Industries
- Polymers, Paints, and Coatings
- Specialty and Fine Chemicals
- Catalysts and Advanced Materials Manufacturing
- Food and Beverage Processing
- Energy and Power Sector (Thermal, Nuclear, and Renewables)
- Environmental and Waste Management Solutions
- Consumer Products and Personal Care
- Data Analytics and Process Optimization

DEPARTMENT
OF
CHEMICAL ENGINEERING



Student Demography



B.Tech

87 Student
4 Year Program Admission
through JEE Basic Engineering
& Departmental Courses
Undergraduate Projects
3rd year B.Tech Eligible for 6
month internship

M.Tech & MS(R)

33 M.Tech Students & 3 MS(R)
2 Year Program
Admission through GATE
Compulsory Departmental
Courses & Electives
1.5 Years Thesis

Dual Degree

03 Student
5 Year Program Admission
through JEE Basic
Engineering &
Departmental Courses 1.5
Years Thesis

PhD

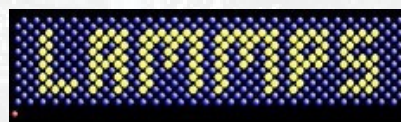
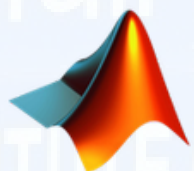
8 Student
Admission through written test
& Interview Compulsory
Departmental Courses &
Electives
PhD Thesis (3+ years)



Skills



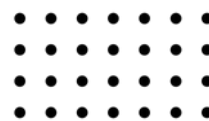
Software Skills:



Laboratory Skills:

- High Resolution Field Emission Scanning Electron Microscope (FESEM) Inductively
- Coupled Plasma-Mass Spectrometer (ICP-MS)
- Automated Gas Adsorption Set up for Physisorption, Chemisorption & Temperature Programmed Oxidation, Reduction & Desorption processes.
- Thermal Analyzers: Thermogravimetry (TGA) & Differential Scanning Calorimetry (DSC)
- Electrochemistry set up with Potentiostat / Galvanostat
- Powder & Thin film X-ray Diffractometer
- Micro Particle Image Velocimetry (MPIV)
- Universal Testing Machine (UTM)
- Electrospinning Unit with 3D Bioprinter as Accessory
- Polarization Microscope with Fluorescence spectroscopy & Peltier Stage attachment
- Advanced Rheometer [MCR 702e, AntonPaar]

Department Activities



Research Scholar Day is a yearly one-day event, which showcases the original research of graduate students in the chemical engineering department. The purpose of this symposium is to facilitate scientific discussion, foster student-faculty interactions, and provide a platform for students to present their work to a wider audience. Eminent researchers are invited to deliver lectures every year as part of the plenary talks for the research scholar.

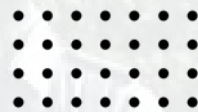


Exergy, the annual chemical engineering fest of IIT Kanpur, embodies the pursuit of purity and perfection before the fall, blending creativity with cutting-edge science. Exergy inspires a transcendental journey through the realms of engineering excellence and exploration.



ChemE Happy Hour, Great conversations, shared stories, and plenty of laughs with our amazing professors and department mates. A fantastic gathering where our PG & UG students connect with professors and fellow department mates in a relaxed setting. From insightful conversations to shared experiences, the event fueling camaraderie and great discussions.

Research Areas



**Materials &
nanotechnology**

**Biotechnology &
Biosystems**



**Catalysis &
Reaction
Engineering**

**Complex Fluids &
Soft Matters**



**Theory,
Computation &
Machine Learning**

**Energy &
Sustainability**



State of the Art Facilities



- Rheometer Polarized Optical
- Micro PIV
- Atomic Force Microscope
- Optical Profilometer
- Real time PCR
- Atomic Absorption Spectroscopy
- Confocal laser scanning microscope
- Dispersive Raman Spectrometer
- Nano Imprint Lithography
- 3D Bioprinter
- ICP Mass Spectrometer
- Surface Area Analyser
- Universal Testing Machine
- Temperature Programmed Reduction
- Continuous Distillation Column

Placement Process

Fill Form: Complete the Job Announcement Form (JAF) / Internship Proforma (IP) on the aforementioned RAS portal.

Selected Students: Companies provide the list of selected and waitlisted students at the end of their interview slot.

Offer Letter: The company sends the offer letter to the selected candidates consistent with JAF or IP.

1

Register: Register as an recruiter by Signing Up on the Recruitment Automation System Portal (RAS) by clicking [here](#)

2

Shortlisted for interview: After reviewing resumes & portfolios - The company submits the list of shortlisted students for the interview process.

3

4

Receive Acceptance: The Placement office notifies the companies regarding the acceptance of selected students.

5

6



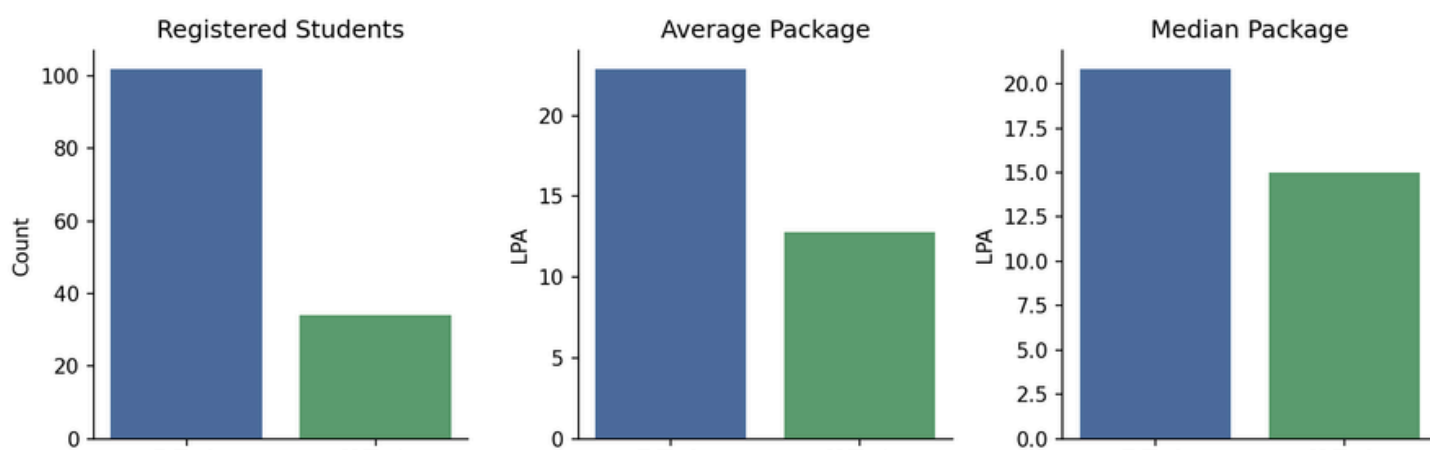
Record-Breaking Innovation Milestone

150 + IPR Filed from 2013

90 + IPR Granted from 2013

150 + Publications per year

Placement Overview Statistics (2025-26)



Program	Registered Students	Average (LPA)	Median (LPA)
BTech	102	22.89	20.85
MTech	34	12.76	15.00

- Around 25% of Total Institute IPRs are from the Chemical Engineering Department in 2024 (Dept. IPR Total No - 38, IITK Total IPR No - 152)
- IITK has been honored with the STEM Impact Awards in 2024 and 2022 for impactful technology transfer from the department, contributing to a broader socio-economic impact.

Notable Alumni



Dr. Ashutosh Sharma
Padma Shree Awardee,
President, Indian
National Science Academy



Mr. Anup Bagchi
CEO, ICICI Pru Life
Insurance



Ms. Vartika Shukla
Chairperson and
Managing Director
Engineers India Limited



Mr. Hemant Jalan
Managing Director
Indigo Paints Ltd.



Dr. Rajeev Gautam
Performance Materials
and Technologies,
Honeywell



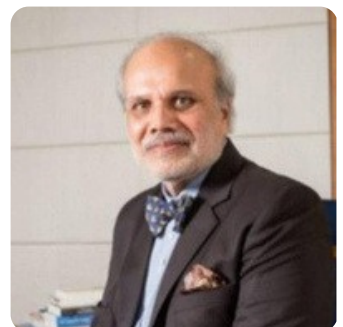
Ms. Neera Tandon
Co-owner,
Telcom Ventures



Mr. Jagjeet Singh Bindra
Former President
Chevron Technology Ventures

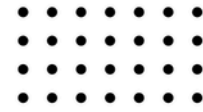


Mr. Rajat Dhawan
Senior Managing Partner,
McKinsey & Company India



Dr. K.K. Sharma
Vice Chairman,
Lupin Limited

Collaborators and Sponsors



Dr.Reddy's



Hindustan Unilever Limited



TATA STEEL



tcs TATA
CONSULTANCY
SERVICES



Balrampur Chini Mills Limited



JSW
Cement



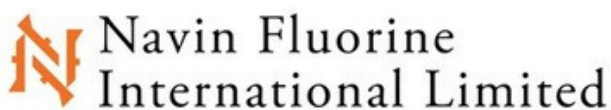
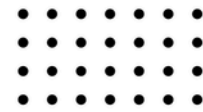
AARTI
INDUSTRIES



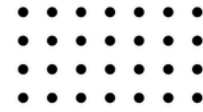
DCM SHRIRAM
Growing with trust

Sterlite

Current Recruiters



Current Recruiters



Coordinators



Faculty Coordinator



Prof. Salman Ahmad Khan



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Student Coordinators



Ayush Verma



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